



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

BIOLOGY
PAPER 1 Multiple Choice

6030/1
1 hour

NOVEMBER 2025 SESSION

Additional materials:
Multiple Choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)
Electronic calculator

INSTRUCTIONS TO CANDIDATES

Write your name, centre number and candidate number on the answer sheet in the spaces provided unless this has already been done for you.

Read very carefully the instructions on the answer sheet.

Answer **all** questions. For each question, there are four possible answers, **A, B, C** and **D**, choose the correct answer. Record your choice in **soft pencil** on the separate answer sheet.

INFORMATION FOR CANDIDATES

There are **forty** questions in this paper. Each correct answer will score **one** mark. Any rough working should be done on this question paper.

This question paper consists of 12 printed pages.

Copyright: Zimbabwe School Examinations Council, N2025.

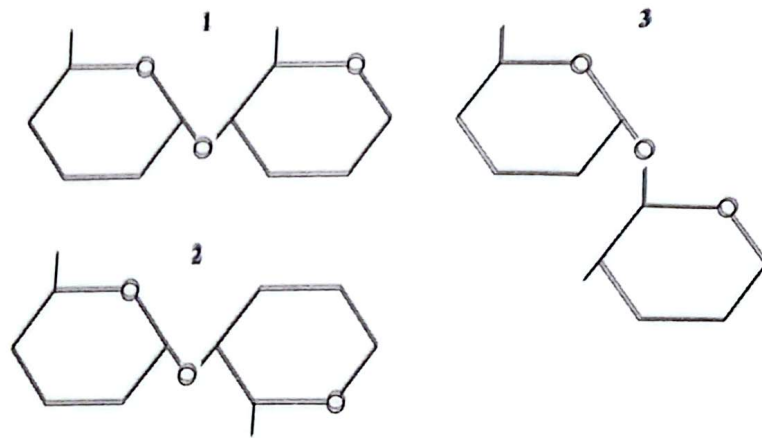


- 1 What is resolution, with reference to microscopy?
 - A The number of times the image of an object appears bigger.
 - B The ability of a microscope to show internal parts of a specimen.
 - C The shortest distance between separate objects that a microscope can show.
 - D The type of radiation used by a microscope to produce an image of a specimen.

- 2 Which one is a function of the rough endoplasmic reticulum?
 - A transports proteins
 - B synthesis of steroids
 - C makes lysosomes
 - D internal processing

- 3 Which component of the cell surface membrane is responsible for cell-cell recognition?
 - A cholesterol
 - B glycolipids
 - C phospholipids
 - D enzymes

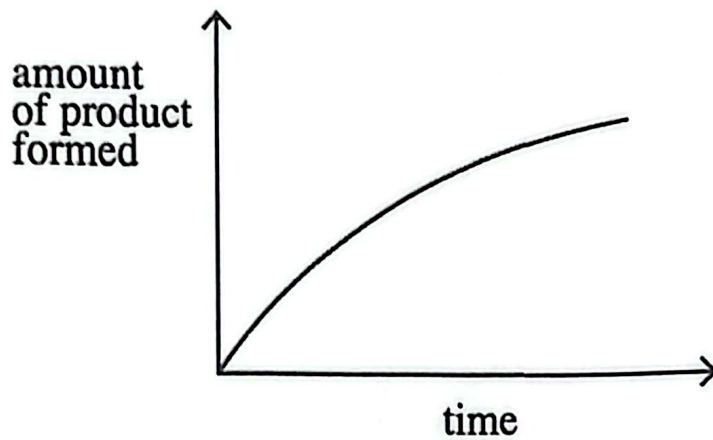
- 4 The diagrams show different types of bonds found in polysaccharides.



Which type(s) of bonds is/are found in amylose?

- A 1 only
- B 2 only
- C 1 and 3
- D 2 and 3

- 5 The graph shows the progress of starch digestion by the enzyme salivary amylase.

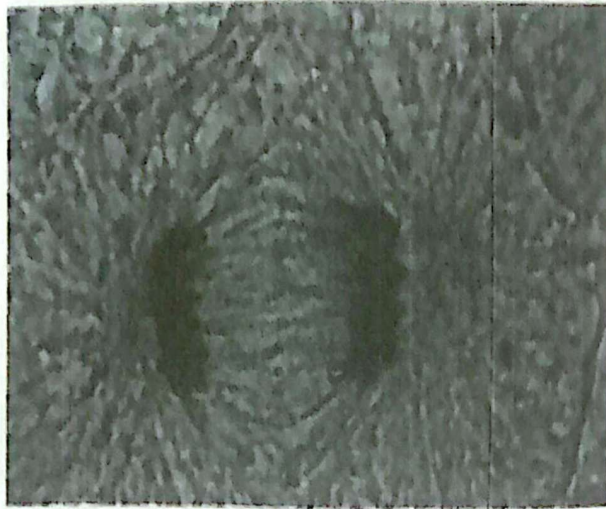


Why does the reaction slow down after some time?

- A end product inhibition by maltose
 - B salivary amylase is becoming denatured
 - C salivary amylase is becoming saturated with starch
 - D the number of starch molecules is decreasing
- 6 Which row correctly describes a triglyceride?
(✓ = yes × = no)

	soluble in water	provides energy	produces water when oxidised
A	×	✓	✓
B	×	✓	×
C	✓	×	✓
D	✓	✓	×

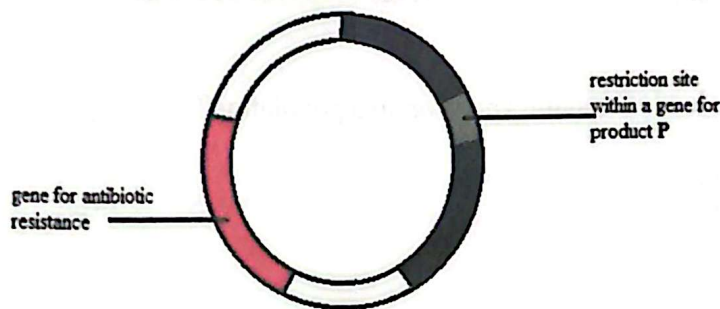
- 7 The photomicrograph shows a stage in nuclear division.



Which stage is shown in the photomicrograph?

- A anaphase
 - B interphase
 - C metaphase
 - D telophase
- 8 A parent cell contains a nucleus with diploid number of 8.
How many possible types of gametes can be produced from it as a result of independent assortment?
- A 4
 - B 8
 - C 16
 - D 32
- 9 An analysis of DNA from cells in a human body showed that thymine makes up 15% of the nucleotide bases.
What would be the percentage composition of guanine in the DNA?
- A 15
 - B 35
 - C 70
 - D 85
- 10 Semi conservative replication of DNA means
- A one strand is copied continuously and the other staggered during replication.
 - B the new molecule has alternating new and old bands.
 - C there is a new and old strand in the molecule after replication.
 - D half of the molecules are old and half are new after replication.

- 11 Which one is an earliest stage during protein synthesis?
- A translation of the triplet code on mRNA to amino acids
 - B attachment of the ribosome to mRNA
 - C amino acid activation and attachment to tRNA
 - D conversion of the DNA triplet code to mRNA
- 12 Which enzyme is **not** needed in the production of recombinant DNA from mRNA?
- A DNA ligase
 - B reverse transcriptase
 - C restriction endonuclease
 - D RNA polymerase
- 13 Which one is **not** a benefit of genetic fingerprinting?
- A Reducing incidence of violent crime.
 - B Solving parental disputes and cases of incest.
 - C Reducing inbreeding of endangered species.
 - D Resolving disputes over family relationships.
- 14 The diagram shows a plasmid that can be used as a vector for transforming bacterial cells in the production of a biological product. The plasmid has two marker genes.



- Which statement correctly describes the use of the marker genes in identifying transformed cells.
- A Transformed cells are resistant to the antibiotic.
 - B Transformed cells produce product P.
 - C Transformed cells are not resistant to the antibiotic.
 - D Transformed cells will not produce product P.
- 15 A child is diagnosed with a rare genetic disease. Neither parent has the disease. How might the child have inherited the disorder?
- A The disorder is due to an environmental factor.
 - B The disorder is dominant and was carried by a parent.
 - C The disorder is sex linked and inherited only from the father.
 - D The disorder is recessive and carried by both parents.

16 A trait controlled by four alleles is said to have

- A incomplete dominance.
- B complete dominance.
- C homologous alleles.
- D multiple alleles.

17 In garden peas, *Pisium sativum*, round seed is dominant over wrinkled seed and yellow cotyledon is dominant over white cotyledon.

In a cross between heterozygous plants for both traits, what is the probability of producing wrinkled seeds with yellow cotyledons?

- A 0.0625
- B 0.1875
- C 0.5625
- D 0.7500

18 The following are some of the causes of variation in a population.

1. crossing over during meiosis
2. different environmental conditions
3. random mating
4. mutation

Which of these lead to genetic variation in a population?

- A 1, 2 and 4
- B 1, 2 and 3
- C 1, 3 and 4
- D 2, 3 and 4

19 Some events that may occur within a population and its environment are:

1. gene mutation
2. selection pressure
3. predation of prey
4. environmental shift
5. change of seasons

Which events are involved in the formation of a new species?

- A 1, 2 and 3
- B 1, 2 and 4
- C 2, 3 and 5
- D 3, 4 and 5

- 20 Antimycin is an insecticide which inhibits the electron transport chain.

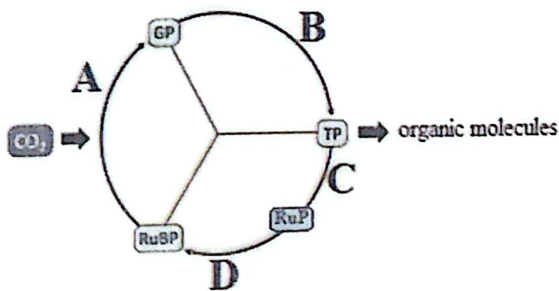
Which row correctly shows the effect of antimycin?

	lactic acid	oxygen consumption	Krebs cycle
A	increases	increases	decreases
B	decreases	decreases	increases
C	increases	decreases	decreases
D	decreases	increases	decreases

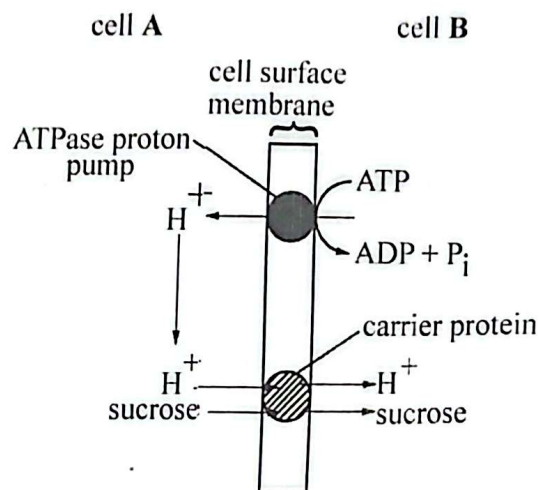
- 21 What causes the respiratory quotient (RQ) of germinating seeds to decrease from 1.0 to 0.7 with time?

- A the respiratory substrate changes from carbohydrate to lipid
- B the respiratory substrate changes from lipid to carbohydrate
- C aerobic respiration replaces anaerobic respiration
- D anaerobic respiration replaces aerobic respiration

- 22 The diagram shows the Calvin Cycle.
Which step, A, B, C or D, uses both ATP and NADPH₂?



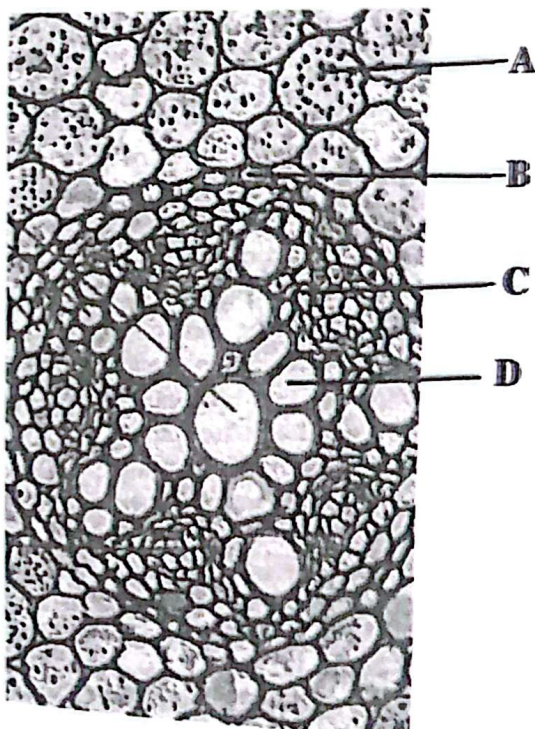
- 23 The diagram represents the loading of sucrose.



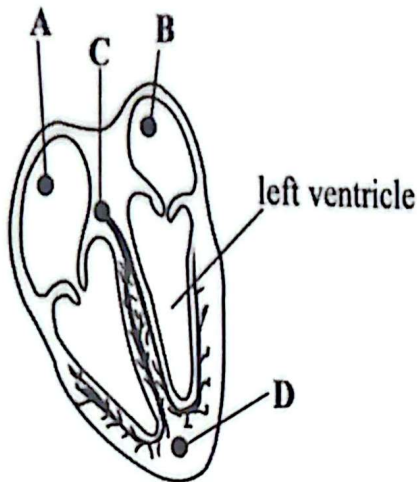
What are the identities of cell A and cell B?

	cell A	cell B
A	mesophyll	sieve tube element
B	mesophyll	companion
C	companion	sieve tube element
D	companion	mesophyll

- 24 The micrograph shows a section through a root.
Which labelled tissue contains the Casparian strip?

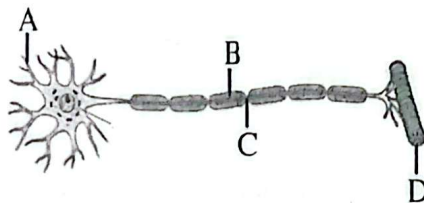


- 25 The diagram below shows the section through the human heart.
Where is the atrioventricular node located?



- 26 Which statement best defines systolic blood pressure?
- A the maximum blood pressure in the right ventricle
 - B the maximum blood pressure in the arteries
 - C the blood pressure in the arteries
 - D the blood pressure in the left ventricle at the end of contraction
- 27 Which of the following statements, about hyper-polarisation, is correct?
- A It is caused by the generation of an action potential.
 - B It is a result of a delay in the closing of sodium gates.
 - C It is because of a delay in the closing of potassium gates.
 - D It is caused by the depolarisation of the excited membrane.
- 28 The diagram shows a motor neurone.

Which part, A, B, C or D, allows the formation of a local circuit?

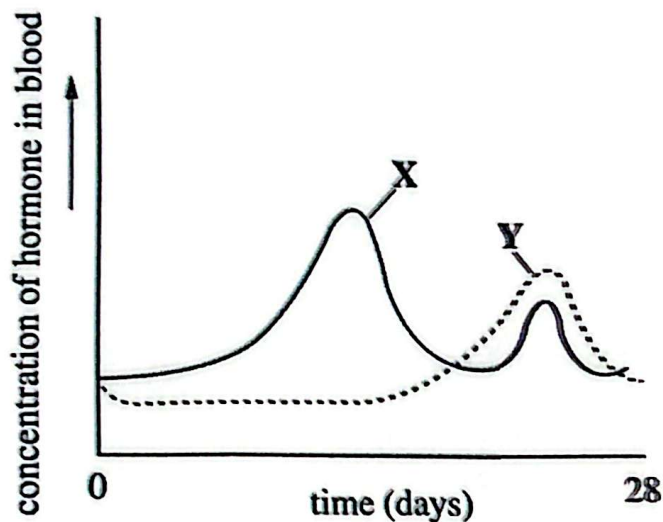


- 29 Which enzyme, secreted by a sperm, helps in its penetration into the ovum?
- A amylase
 - B catalase
 - C invertase
 - D protease

30 Which hormone is **not** secreted by the placenta during pregnancy?

- A chorionic gonadotrophin
- B luteinising hormone
- C oestrogen
- D progesterone

31 The graph shows changes in the concentration of hormones X and Y in the blood during the menstrual cycle.



Which of the following correctly identifies hormones X and Y?

	hormone X	hormone Y
A	LH	oestrogen
B	oestrogen	FSH
C	oestrogen	progesterone
D	progesterone	oestrogen

32 Release of untreated sewage into rivers decreases

- A carbon dioxide concentration.
- B nitrate concentration.
- C phosphate concentration.
- D oxygen concentration.

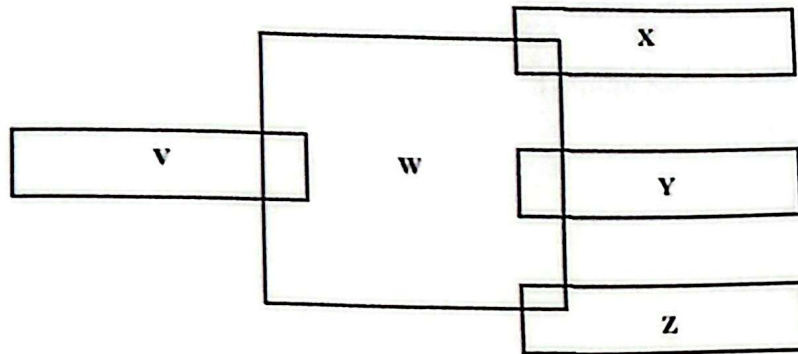
33 Which agency or organisation in Zimbabwe directly involves the community in ecological conservation efforts?

- A CAMPFIRE
- B UNESCO
- C EMA
- D WWF

34 Which one is a diagnostic feature of organisms in the phylum Cnidaria?

- A they are triploblastic
- B possess radial symmetry
- C exist as polyps or medusae
- D exhibit polymorphism

35 The diagram shows one way of classifying organisms into the five kingdoms.



Which kingdom is represented by the box labelled W?

- A fungi
- B plantae
- C prokaryotae
- D protista

36 Which taxon has the **lowest** diversity of living organisms?

- A genus
- B phylum
- C order
- D family

37 Which condition explains the high prevalence of malaria in the tropical regions?

- A high temperature
- B high wind speed
- C low rainfall
- D low humidity

- 38 The table shows information on diseases and factors affecting distribution. Which row is correctly matched?

	disease	factor affecting distribution
A	coronary heart disease	consumption of unsaturated fats
B	Ebola	human interaction with primates
C	HIV/AIDS	poor sanitary conditions
D	malaria	overcrowding

- 39 Which disease is linked to a sedentary lifestyle?

- A cholera
- B Ebola
- C sickle cell anaemia
- D coronary heart disease

- 40 Which disease has the same global distribution as malaria?

- A cholera
- B coronary heart disease
- C sickle cell anaemia
- D ebola